

Spreading 42 Points in the Plane

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Abstract

We place 42 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 39.71826+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 43-point configuration gives 40.63406; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 42 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 39.71826\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 42$ is not on the page. Deleting a point from our 43-point configuration gives 40.63406; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n42.txt
```

in <https://github.com/Ge-limin/math719>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
39718260236823674496400313
denominator:
999999999998115725761970
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

42 points, x and y per line. File: friedman-maxmin2d/coords/n42.txt.

```
-2.020742892317 -4.998631242118 -3.973127788384 -4.669758249899
-5.286004191788 -1.124734218641 0.296733250745 -1.668185409512
-5.133897595883 -3.071946484902 -2.257601639453 -0.723156794926
-3.461634606328 -1.928737055421 -0.265239931602 -0.841029957181
0.170111111822 0.059230821144 0.732084294170 -0.767924631188
-3.511131475333 1.096254394309 -1.062985203979 -4.711054799360
-1.822934444440 0.177434360693 -4.280684495077 -0.870991979328
-1.367375033634 -2.489024153100 -0.959211943927 -3.401933180441
0.425689087466 -3.387475356499 -1.679136109338 -1.538863623676
-3.686945294666 -0.066334436087 -4.458910954427 -1.854981475811
-5.424487938703 -2.115098925121 -2.363850626073 1.294264539623
-3.230139636723 -0.955900954607 -3.715607615315 -3.703485327551
-2.171129836618 -3.083984838087 -0.700392415331 -1.743950876917
0.866598346317 -2.489923672495 -1.387583401016 1.077695139018
-4.681184599592 -3.963602776861 -0.217494955433 -2.621763735205
-2.762911453993 -4.007409700055 -1.762966746911 -3.996893865428
-3.149873530625 -2.878897584846 -1.244468914325 -0.638272468056
-4.992614300913 -0.168741346318 -0.259230400995 -4.116094114373
-4.168320611606 -2.811829035592 -2.799201669497 0.394003761299
-3.020431627062 -4.973682622403 -0.809117870901 0.261988310268
-4.398875100502 0.635916196923 -2.482890912322 -2.133824308663
```